WORLD HEALTH ORGANIZATION (WHO)

Hypertension Factsheet & Global Cardiovascular Epidemiology

Arabic Reference: Sahifat Waqa'i Fart Daght Al-Dam (WHO Factsheet No. 342)

Official Reference: WHO Factsheet 342 / Global NCD Monitoring Framework

Official Web Portal: https://www.who.int/ar/news-room/fact-sheets/detail/hypertension

1. Global Disease Burden & Epidemiology (Key Facts)

* An estimated 1.4 billion adults aged 30-79 years worldwide live with hypertension.
* Two-thirds of people with hypertension live in low- and middle-income countries (LMICs).
* Approximately 46% of adults with hypertension are completely unaware that they have the condition.
* Only 42% of adults with hypertension are diagnosed and treated, and only 21% have it controlled.
* Hypertension is a major independent cause of premature death globally, causing 10.8 million deaths annuall

2. The 'Silent Killer' Clinical Concept & Pathophysiology

* Hypertension is medically termed 'The Silent Killer' because it typically produces zero symptoms for years
* Insidious hydraulic shear stress progressively damages the vascular endothelial lining across vital organs
* Absence of symptoms does not indicate absence of disease; relying on feelings or mild headaches is medical
* Routine annual blood pressure screening is universally mandatory for all adults >=18 years.

3. Modifiable and Non-Modifiable Cardiovascular Risk Factors

A. Modifiable Behavioral Risk Factors:

* Excessive dietary sodium consumption (>2,300 mg/day) and deficient potassium intake (<3,500 mg/day).
* Physical inactivity, sedentary lifestyle, and high prevalence of abdominal obesity (BMI >=30).
* Consumption of tobacco products (cigarettes, shisha/hookah) and harmful alcohol intake.
* Diets high in saturated fats, trans-fatty acids, and refined sugars.

B. Non-Modifiable Risk Factors:

* Advanced age (>=65 years) leading to physiological vascular elastance reduction and large artery stiffness
* Genetic predisposition and positive family history of early cardiovascular disease.
* Co-existing chronic conditions: Diabetes mellitus, chronic kidney disease (CKD), and metabolic syndrome.

4. Severe Hypertensive Crisis & Target Organ Warning Signs

* Readings persistently exceeding 180 mmHg systolic and/or 120 mmHg diastolic constitute an emergency.
* Critical danger signs signaling acute Target Organ Damage (TOD) include:

1. Sudden, severe 'thunderclap' headache.
2. Acute crushing chest pain, tightness, or pressure (angina pectoris / acute coronary syndrome).
3. Acute severe dyspnea, shortness of breath, or orthopnea (pulmonary edema).
4. Sudden focal neurological deficit: Facial droop, hemiparesis, dysarthria, or numbness.

5. Acute visual disturbance, blurry vision, or sudden loss of vision (retinopathy / papilledema).
6. Severe vertigo, altered mental status, confusion, or seizures (hypertensive encephalopathy).
   * Mandatory Immediate Action: Call emergency services (997 / 122 / 911) or proceed to nearest emergency room

5. Chronic Target Organ Complications & Long-Term Prognosis

* Cardiac: Left ventricular hypertrophy (LVH), coronary artery disease, heart failure with preserved/reduced
* Cerebrovascular: Ischemic stroke, intracerebral hemorrhage, transient ischemic attack (TIA), vascular deme
* Renal: Nephrosclerosis, microalbuminuria, progressive CKD, end-stage renal disease (ESRD) requiring dialys
* Ocular: Grade I-IV hypertensive retinopathy, flame hemorrhages, cotton wool spots, permanent blindness.

6. Universal Blood Pressure Treatment Targets

* General Target: Blood pressure <140/90 mmHg for all confirmed adult hypertension patients.
* Intensive Target: Blood pressure <130/80 mmHg for patients with confirmed CVD, diabetes, or CKD.